EV-GUARDIAN 2.0:
INTELLIGENT EV BATTERY SAFETY, PREDICTIVE DIAGNOSTICS AND SAFE-DRIVE INTERLOCK

A MINI PROJECT REPORT

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ABSTRACT

Battery safety is one of the most critical requirements for the wider adoption of electric vehicles (EVs). A conventional battery management system (BMS) protects the pack with threshold-based limits, but a threshold alarm alone does not tell an operator what is abnormal, where it is occurring, how serious it is, or what should be done next. This project presents EV-GUARDIAN 2.0, a low-cost embedded diagnostic layer that works alongside the BMS of a low-voltage 4S battery demonstrator. An ESP32 edge controller acquires cell voltage, pack current, four-point temperature and cooling / emergency-stop status, validates every signal for range, freshness and plausibility, and extracts features such as cell-voltage imbalance, temperature difference, temperature rise rate and current trend. A hybrid diagnostic engine combines deterministic safety rules with an edge anomaly score to localize the suspected cell or condition and to compute an explainable EV Safety Score between 0 and 100.

The diagnosis drives a closed-loop response consisting of warning, cooling, controlled isolation and event logging to a microSD Safety Black Box, together with a fail-safe Safe-Drive Interlock that blocks the drive / contactor permit whenever a required safety input is missing, invalid or critical. The design is demonstrated through a software-in-the-loop simulator that injects six controlled fault scenarios: cell thermal rise, cell undervoltage, overcurrent, sensor failure, cooling failure and multiple faults. In every injected scenario the simulator lowered the Safety Score, identified the affected location and blocked the drive permit, and the permit was restored only after the fault was cleared and the inputs were revalidated. The estimated budget for the hardware demonstrator is about ₹4,400. All testing is limited to simulated faults on low-voltage, current-limited hardware; destructive lithium-battery testing is explicitly excluded, and the hardware prototype is under development.

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LIST OF ABBREVIATIONS

ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
BMS	Battery Management System
CAD	Computer-Aided Design
CAN	Controller Area Network
E-stop	Emergency Stop
EV	Electric Vehicle
FSM	Finite-State Machine
GPIO	General-Purpose Input / Output
I²C	Inter-Integrated Circuit
IoT	Internet of Things
Li-ion	Lithium-ion
ML	Machine Learning
MUX	Multiplexer
OLED	Organic Light-Emitting Diode
RUL	Remaining Useful Life
SDG	Sustainable Development Goals
SIL	Software-in-the-Loop
SoC	State of Charge
SoH	State of Health
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVE

The main objective of this project is to design and demonstrate **EV-GUARDIAN 2.0**, a low-cost embedded diagnostic layer that continuously watches the electrical and thermal condition of an electric vehicle (EV) battery pack, explains what is abnormal and where it is occurring, and takes a defined protective action, while preventing unsafe activation of the drive or contactor request.

The specific objectives are:

- To acquire cell voltage, pack current, multi-point temperature and cooling / emergency-stop status from a safe, low-voltage 4S battery demonstrator using an ESP32 edge controller.
- To validate every signal for range, freshness and plausibility before it is used in any decision.
- To extract diagnostic features such as cell-voltage imbalance, temperature difference across sensors, temperature rise rate and current trend.
- To combine deterministic safety rules with an edge anomaly score and localize the suspected cell or condition when the evidence supports it.
- To calculate an explainable EV Safety Score between 0 and 100 and to display the reasons behind every reduction in the score.
- To implement closed-loop protection (warning, cooling, controlled isolation) and a fail-safe Safe-Drive Interlock that defaults to “no permit”.
- To record every event in a Safety Black Box so that the sequence of readings, diagnosis and action can be reviewed after the event.
- To validate the complete workflow safely using controlled, simulated faults, without any destructive lithium-battery testing.

1.2 OVERVIEW OF EV BATTERY SAFETY AND INTELLIGENT DIAGNOSTICS

1.2.1 Importance of battery safety in electric vehicles

The lithium-ion battery pack is the most valuable and the most safety-critical component of an electric vehicle. Cells in a pack can drift apart in voltage, heat unevenly, be over-stressed by excessive current, or be monitored by a sensor that has

silently failed or lost its connection. If such conditions are not noticed early, they can lead to reduced battery life, sudden loss of power or, in the worst case, thermal events. Reliable early detection and clear communication of battery risk are therefore essential for user safety, maintenance planning and public confidence in electric mobility.

In practice, the measurable indicators of an abnormal pack are cell-voltage deviation or imbalance, absolute cell temperature, temperature difference across sensors, temperature rise rate, pack current, sensor validity, cooling status and the state of the drive interlock. EV-GUARDIAN 2.0 is built around exactly these indicators.

1.2.2 Limitations of existing approaches

A battery management system (BMS) is essential and is not replaced by this project. However, a simple threshold alarm only reports that a limit has been crossed. It gives limited context about the suspected location, the severity, the likely explanation and the recommended action. Table 1.1 summarizes the common approaches and the gap that EV-GUARDIAN 2.0 addresses.

Table 1.1: Existing approaches and the gap addressed by EV-GUARDIAN 2.0

Approach	Strength	Gap addressed by EV-GUARDIAN 2.0
Basic threshold alarm	Simple and fast	Limited context, localization and explanation
BMS protection	Essential battery monitoring and protection	Project adds a diagnostic / interpretation layer instead of replacing the BMS
Cloud monitoring	Remote visibility	May add latency and network dependency; this project keeps the core diagnostic logic at the edge
AI anomaly detection alone	Can detect complex patterns	Safety-critical actions must not depend only on machine learning; deterministic interlocks remain

A further safety gap is the unintended activation of the drive or contactor when the battery, a required sensor or the emergency stop is not in a healthy state. EV-GUARDIAN 2.0 closes this gap with an explicit, fail-safe interlock.

1.2.3 The EV-GUARDIAN 2.0 concept

EV-GUARDIAN 2.0 is a low-cost intelligent diagnostic layer that sits beside the BMS. It follows a seven-step philosophy: **SENSE → DIAGNOSE → LOCALIZE → PREDICT → PROTECT → INTERLOCK → RECORD**. Instead of reporting only

“limit crossed”, the system answers four practical questions: what is abnormal, where it is likely occurring, how serious it is, and what should happen next. A dynamic EV Safety Score, an explanation panel, protective outputs and a Safety Black Box make the result easy to understand and review. The prototype is a visible, transparent low-voltage demonstrator so that the relationship between the battery, the sensors, the controller and the protection hardware can be seen directly.

1.3 DIRECT BENEFITS

- **Multi-parameter diagnosis:** voltage, current, temperature and sensor health are correlated instead of relying on a single alarm.
- **Fault localization:** a suspected cell or condition is identified when the available measurements support it.
- **Explainable safety score:** a single 0–100 indicator is shown together with the reasons that reduce it.
- **Closed-loop response:** diagnosis is connected to warning, cooling, controlled isolation and event logging.
- **Accidental-activation prevention:** the Safe-Drive Interlock blocks the drive / contactor request whenever a required input is missing, invalid or critical.
- **Post-event review:** the Safety Black Box stores sensor history and event context.
- **Low cost:** the demonstrator uses widely available modules with an estimated budget of about ₹4,400.

1.4 INDIRECT BENEFITS

- Supports safer maintenance and faster fault review for battery operators and maintenance teams.
- Provides a low-cost educational platform for teaching embedded EV battery safety.
- Creates a modular path towards CAN integration, fleet analytics and digital-twin models.
- Encourages safer adoption of e-scooters, cars, buses and charging infrastructure, supporting the wider goals of clean and sustainable transportation.

1.5 ORGANIZATION OF THE REPORT

The report is organized into five chapters. **Chapter 1** states the objective, gives an overview of EV battery safety and lists the benefits. **Chapter 2** reviews the related literature and identifies the research gap. **Chapter 3** describes the system implementation, including the architecture, hardware, software, diagnostic algorithms, safety score, interlock and the software-in-the-loop simulator. **Chapter 4** presents the results obtained from the simulated fault scenarios. **Chapter 5** concludes the work and outlines future trends. The appendix gives core pseudocode, and the references are listed at the end.

CHAPTER 2
REVIEW OF LITERATURE

2.1 RELATED BACKGROUNDS

This chapter summarizes the background topics on which EV-GUARDIAN 2.0 is built: battery hazards, battery management systems, fault diagnosis and anomaly detection, edge computing, and functional safety. The broad technologies behind battery monitoring, BMS protection, AI anomaly detection and fault diagnosis already exist in research and industry; the contribution of this project is therefore the integrated, low-cost embedded implementation and the way the modules are demonstrated together.

2.1.1 Lithium-ion battery hazards and thermal runaway

Feng et al. [4] reviewed the thermal runaway mechanism of lithium-ion batteries for electric vehicles, describing how abuse conditions such as overheating, overcharge or mechanical and electrical stress can trigger a chain of self-heating reactions inside a cell and how the failure can spread to neighbouring cells. The review highlights why early detection of abnormal temperature behaviour, rather than only a high absolute temperature, is valuable. This motivates the use of temperature rise rate and temperature difference across sensors as diagnostic features in EV-GUARDIAN 2.0.

2.1.2 Battery management systems

Lu et al. [3] reviewed the key issues in lithium-ion battery management for electric vehicles, including state estimation, cell balancing, thermal management and safety protection. Rahimi-Eichi et al. [5] presented an overview of battery management systems for electric vehicles and smart-grid applications, and Hannan et al. [6] discussed the state of the art and the issues and recommendations for energy management of lithium-ion batteries in EV applications. Plett's textbooks [1], [2] provide the modelling and estimation foundation used in modern BMS design. These works establish the BMS as the essential protection layer; they also show that BMS research concentrates mainly on state estimation, balancing and protection limits. EV-GUARDIAN 2.0 therefore does not replace the BMS, but adds an interpretation layer that explains risk and records the event.

2.1.3 Fault diagnosis and anomaly detection

Battery fault diagnosis is commonly approached through threshold and rule-based methods, model-based methods and data-driven or machine-learning methods. Rule-

based methods are simple, fast and predictable, but they may give limited insight when several signals drift together. Data-driven anomaly detection can capture complex patterns, but it depends on the quality and coverage of training data and may behave unpredictably in situations that were not represented during training. For this reason, EV-GUARDIAN 2.0 uses a hybrid design: deterministic rules form the baseline, a lightweight anomaly score enhances detection, and machine learning is never the sole safety mechanism.

2.1.4 Embedded IoT and edge analytics

Low-cost microcontrollers with integrated Wi-Fi, such as the ESP32 [8], make it possible to acquire sensor data, run filtering and decision logic, and host a local web dashboard on a single board. Sensors suitable for a student demonstrator include the INA219 current and power monitor with an I²C interface [9] and the DS18B20 digital thermometer with a 1-Wire interface [10]. Keeping the core diagnostic logic at the edge avoids dependence on network latency and availability, while an optional cloud connection can be added later for fleet-level analytics.

2.1.5 Functional safety and fail-safe interlocks

ISO 26262 [7] defines the functional-safety lifecycle for road-vehicle electrical and electronic systems and is built on the idea that a system should be able to reach a defined safe state when a fault is detected. The Safe-Drive Interlock of EV-GUARDIAN 2.0 follows this principle at demonstrator level: when a required safety input is missing, invalid or critical, the default state is “no drive / contactor permit”. The project does not claim compliance with ISO 26262; production deployment would require formal automotive functional-safety engineering.

2.1.6 Summary and identified gap

Table 2.1: Summary of the reviewed background and its relation to this project

Reference	Topic	Relevance to EV-GUARDIAN 2.0
Feng et al. [4]	Thermal runaway mechanism in EV lithium-ion batteries	Motivates temperature rise rate and ΔT features for early warning
Lu et al. [3]	Key issues in EV battery management	Defines the BMS functions that the project complements
Rahimi-Eichi et al. [5]	BMS overview for EVs and smart grid	Confirms BMS as the essential protection layer
Hannan et al. [6]	Energy management of Li-ion batteries in EVs	Context for monitoring, issues and recommendations

Reference	Topic	Relevance to EV-GUARDIAN 2.0
Plett [1], [2]	BMS modelling and estimation	Foundation for future SoC / SoH / RUL extensions
ISO 26262 [7]	Automotive functional safety	Basis for the fail-safe, deterministic interlock concept
ESP32 / INA219 / DS18B20 datasheets [8]–[10]	Embedded controller and sensors	Hardware interfaces used in the prototype

From the review, the gap can be stated as follows: existing practice provides essential protection and powerful analytics, but there is room for a low-cost, edge-first layer that links multi-parameter diagnosis, fault localization, an explainable score, a protective response, a fail-safe drive interlock and an event record in a single demonstrable system. EV-GUARDIAN 2.0 is designed to fill this gap. No “first-ever” or patent claim is made, and the integrated workflow must be validated as one system.

CHAPTER 3
SYSTEM IMPLEMENTATION

3.1 DESIGN PHILOSOPHY AND SAFETY BOUNDARIES

The system is designed around three principles. First, **safety comes before intelligence**: deterministic rules and the interlock finite-state machine remain independent of the optional anomaly layer. Second, **explainability**: every score reduction and every blocked permit must be accompanied by a human-readable reason. Third, **fail-safe default**: whenever a required input is missing, invalid or critical, the output defaults to “no permit”.

To keep the demonstration safe, a visible 4S low-voltage battery demonstrator with current-limited hardware is used. All abnormal conditions are created with controlled test signals or software-injected faults. Thermal runaway, intentional short circuits, punctures, uncontrolled overcurrent and other destructive lithium-battery tests are explicitly excluded.

3.2 SYSTEM ARCHITECTURE

The prototype is a transparent enclosure that exposes the battery, sensors, controller, display and protection hardware. Figure 3.1 shows the concept render of the demonstrator and Figure 3.2 shows its exploded view with the main modules labelled. These are design representations of the prototype under development and are not evidence of physical testing.

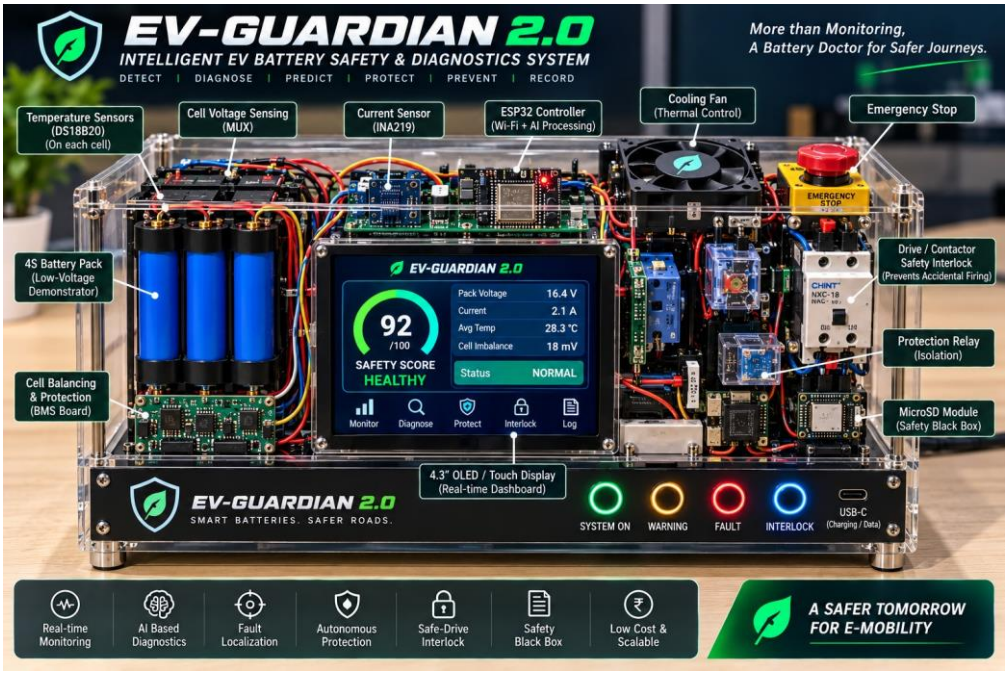


Figure 3.1: Concept render of the EV-GUARDIAN 2.0 demonstrator (prototype under development)

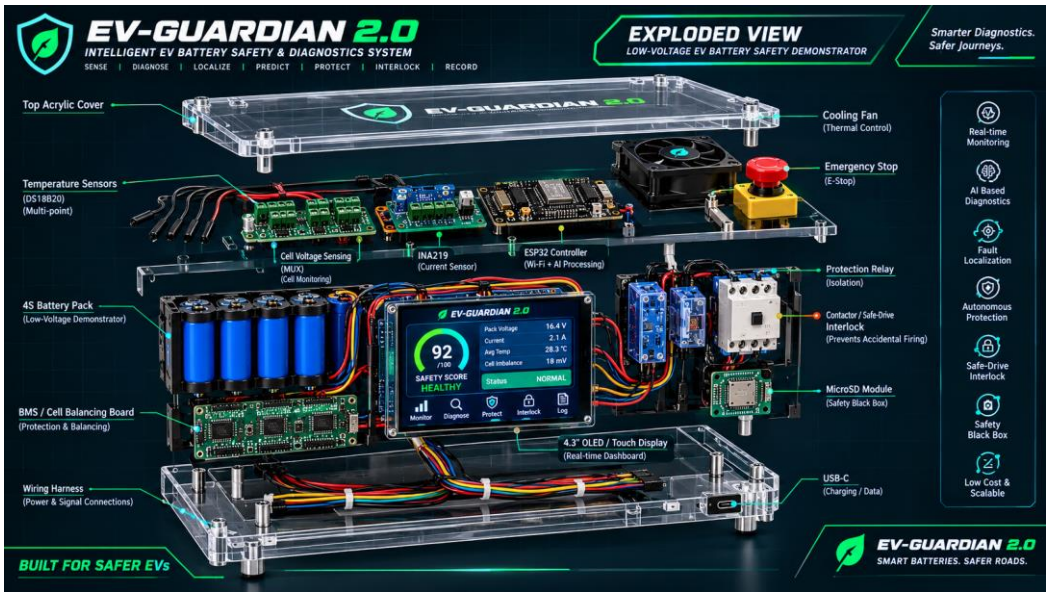


Figure 3.2: Exploded view of the EV-GUARDIAN 2.0 low-voltage demonstrator

The functional architecture is shown in Figure 3.3. Sensors feed the ESP32 edge controller, which performs signal validation and feature extraction. The rule engine and the anomaly score produce the fault localization, the Safety Score and the explanation. The decision then drives four outputs: warning (buzzer and LEDs), cooling (fan), protection (relay) and the Safe-Drive Interlock. All results are also shown on the OLED display and the web dashboard and are stored in the microSD Safety Black Box.

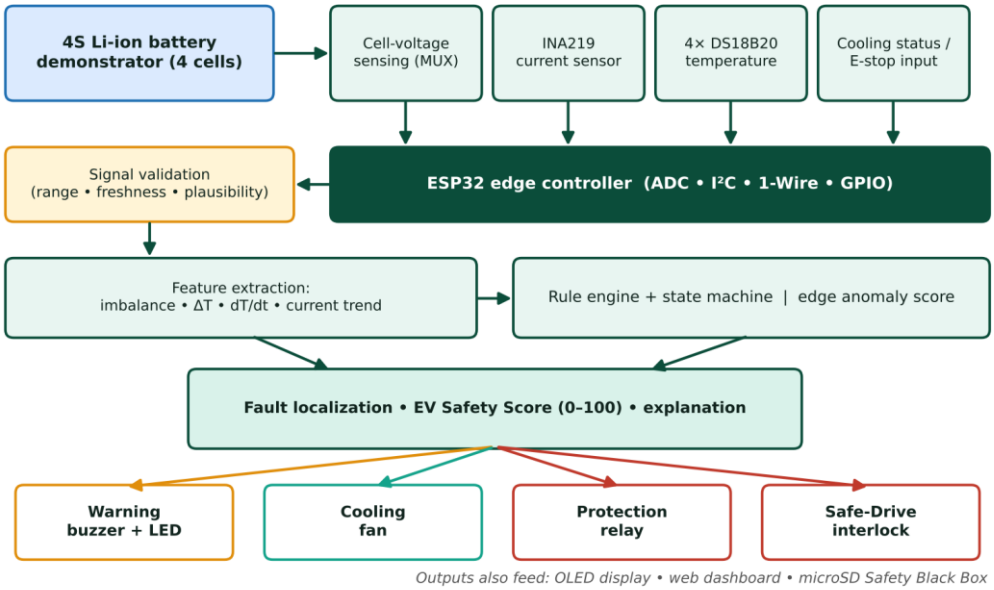


Figure 3.3: Functional architecture of EV-GUARDIAN 2.0

3.3 HARDWARE COMPONENTS

The hardware is built from widely available, low-cost modules. Table 3.1 lists the components, their purpose and the estimated cost. The final bill of materials must be replaced with actual purchase prices when the components are procured.

Table 3.1: Hardware components and estimated cost

Component	Purpose	Est. cost (₹)
ESP32 DevKit	Main controller, Wi-Fi and sensor acquisition	500
4S cells + BMS board	Safe low-voltage demonstrator pack with cell balancing and protection	750
4× DS18B20 sensors	Multi-point temperature monitoring	400
INA219 module	Pack current measurement over I²C	120
Cell-voltage sensing (MUX)	Individual cell voltage measurement	150
OLED / touch display	Local safety dashboard and Safety Score	150
Relay, 5 V fan, buzzer and LEDs	Protection / isolation output, thermal demonstration, alerts	380
microSD module + emergency stop	Safety Black Box logging and manual safe shutdown	350
Enclosure, wiring, connectors	Mechanical and electrical integration	1,000
Itemised total		**3,800**
Miscellaneous allowance (spares, connectors, shipping)	To be confirmed with actual purchase prices	600
Estimated budget build	Showcase version: approximately ₹6,000–₹7,000	**≈ 4,400**

An optional MPU6050 motion sensor may be added for vibration trend sensing. The emergency-stop push button is wired as a hard-wired input so that it can force a safe state independently of the software.

3.4 SENSOR INTERFACING

- **Cell voltage:** the four cell voltages are scaled by a divider / multiplexer stage and read through the ESP32 ADC; the BMS board provides cell protection and balancing.
- **Pack current:** the INA219 measures pack current through a shunt and reports it over the I²C bus.
- **Temperature:** four DS18B20 sensors share one 1-Wire bus, one sensor on each cell, giving four-point thermal monitoring.
- **Cooling and emergency stop:** the fan feedback / status and the emergency-stop contact are read as digital GPIO inputs.
- **Outputs:** the buzzer, LEDs, cooling fan, protection relay and drive-permit output are driven from GPIO pins through a relay / MOSFET stage; the OLED display and microSD module are connected over their serial interfaces.

3.5 SOFTWARE STACK

Table 3.2: Software stack and diagnostic logic

Layer	Implementation
Firmware	ESP32 C/C++ with the Arduino framework
Acquisition	ADC, I²C, 1-Wire and GPIO interfaces
Filtering	Moving average / exponential smoothing with range checks
Safety rules	Thresholds, hysteresis, multi-sensor correlation and a state machine
Anomaly layer	Lightweight edge anomaly score; optional enhancement added after the rule-based baseline
Dashboard	ESP32 local web UI; optional cloud extension
Logging	microSD event log with timestamp, sensor values, score, diagnosis and action
Interlock	Deterministic finite-state machine with a fail-safe no-permit default

3.6 WORKING LOGIC

The firmware runs a repeating cycle, shown as a flowchart in Figure 3.4. In each cycle, the controller reads all inputs, validates them, extracts features, evaluates the safety

rules and anomaly score, localizes the suspected cell or condition and calculates the Safety Score. If a critical fault exists or any interlock input is invalid, the drive permit is blocked and an alarm, protective output and event log are generated. Otherwise the permit is allowed only when all required conditions are healthy. The diagnosis, reason and recommendation are then displayed and the cycle continues.

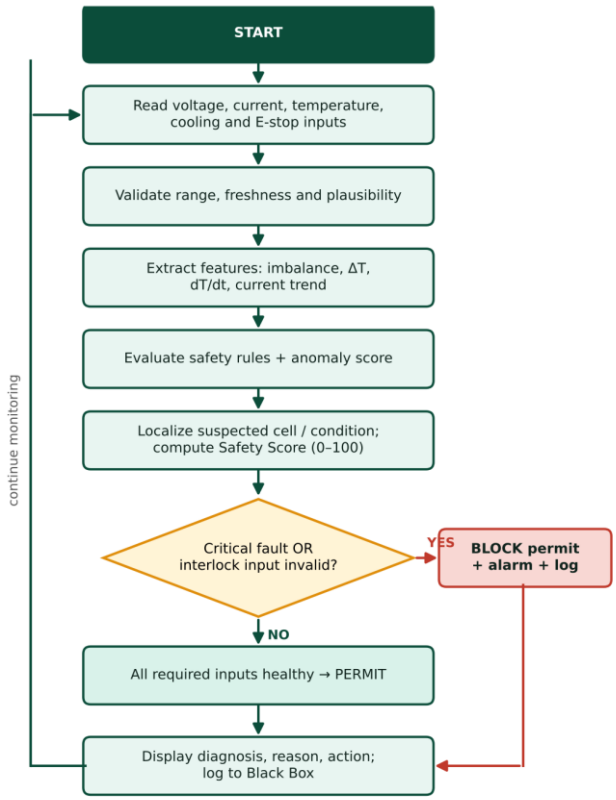


Figure 3.4: Decision flow (working logic) of EV-GUARDIAN 2.0

3.7 SIGNAL VALIDATION

Before any value is used, the following checks are applied. A reading that fails any check is flagged invalid, and if the signal is a required safety input the interlock moves to its fail-safe state.

- **Range check:** the value must lie inside the physically possible range of the sensor.
- **Freshness check:** the sample must have been updated within the allowed time window; missing or stale data is invalid.
- **Plausibility check:** the value must be consistent with related signals and with its own recent history (for example, a temperature that jumps by an impossible amount between two samples).

3.8 FEATURE EXTRACTION

From the validated and filtered signals, the following features are computed in every cycle:

$$\Delta V_{cell} = V_{cell,max} - V_{cell,min} \qquad \textit{(cell-voltage imbalance)}$$

$$\Delta T = T_{max} - T_{min} \qquad \textit{(temperature difference across sensors)}$$

$$dT/dt = (T_k - T_{k-n}) / (n \cdot \Delta t) \qquad \textit{(temperature rise rate)}$$

In addition, the current trend and the consistency between redundant signals are evaluated. The cell with the largest voltage deviation from the pack mean, or with the highest temperature and rise rate, becomes the suspected location.

3.9 RULE ENGINE AND SEVERITY CLASSIFICATION

The rule engine compares the features with configurable limits, using hysteresis so that the state does not chatter when a value hovers around a limit, and uses multi-sensor correlation so that a single noisy sensor does not by itself cause a critical decision. Table 3.3 gives indicative settings. These values are illustrative and must be calibrated for the actual cells during hardware testing.

Table 3.3: Indicative rule settings (to be calibrated on hardware)

Parameter	Monitor	Warning	Critical
Cell-voltage imbalance ΔV_{cell}	> 30 mV	> 60 mV	> 100 mV
Maximum cell temperature	> 37 °C	> 42 °C	> 50 °C
Temperature rise rate	Above normal trend	High	Very high with rising ΔT
Pack current	> 12 A	> 15 A	> 25 A
Sensor validity	—	Optional sensor invalid	Required sensor invalid → fail-safe
Cooling status	—	Cooling requested	Cooling failed while temperature rising

The final severity class is one of **NORMAL, MONITOR, WARNING** or **CRITICAL**, taken as the more severe of the rule result and the score band described in the next section.

3.10 EV SAFETY SCORE

The EV Safety Score is a prototype engineering indicator, not a certified automotive safety metric. It combines normalized risk contributions from voltage imbalance,

thermal deviation, temperature rise rate, current abnormality, sensor validity and other enabled signals:

$$Risk = w_V \cdot R_V + w_T \cdot R_T + w_{\Delta T} \cdot R_{\Delta T} + w_I \cdot R_I + w_S \cdot R_S + \ldots$$

$$Safety\ Score = clamp(100 - Risk,\ 0,\ 100)$$

Each R term is normalized between 0 and its maximum contribution, and the weights w must be tuned and validated during testing. The dashboard shows not only the score but also the top reasons for the reduction, for example “Cell 3 temperature rising” or “temperature rise rate abnormal”. In the dashboard the score is coloured green above 70, amber between 40 and 70, and red below 40; these bands are used as the NORMAL / MONITOR, WARNING and CRITICAL ranges in this prototype.

3.11 FAULT LOCALIZATION

When the available measurements support it, the system names the suspected cell or condition. A single-cell thermal rise with stable neighbours points to that cell; a voltage deviation on one cell points to that cell; a current anomaly with normal cell values points to the pack or load; an implausible or missing reading points to the sensor network; and a rising temperature with unavailable cooling feedback points to the cooling subsystem. Fault localization depends on sensor placement and measurement quality, and its accuracy must be verified for the abnormal conditions represented during validation.

3.12 SAFE-DRIVE INTERLOCK

The Safe-Drive Interlock prevents unintended activation of the demonstrator’s drive / contactor request when critical safety conditions are not satisfied. It is demonstrated only with a low-voltage relay, LED or small DC-load interface, and never by controlling a real vehicle propulsion system. The interlock principle is: **fail-safe default = no drive / contactor permit** whenever a required safety input is missing, invalid or critical. Table 3.4 lists the checks and Figure 3.5 shows the state machine.

Table 3.4: Safe-Drive Interlock checks

Safety check	Required state	If invalid
Battery voltage	Within the configured safe range	Block permit
Temperature	Within the configured range and no abnormal rise trend	Block + warning / cooling
Current	Within the configured test range	Block + warning

Safety check	Required state	If invalid
Sensor validity	All required sensors plausible and available	Fail-safe block
Emergency stop	Released	Block
Critical diagnostic state	No critical fault	Block + alarm + log
Charger / drive request logic	Valid state for the demonstration	Block

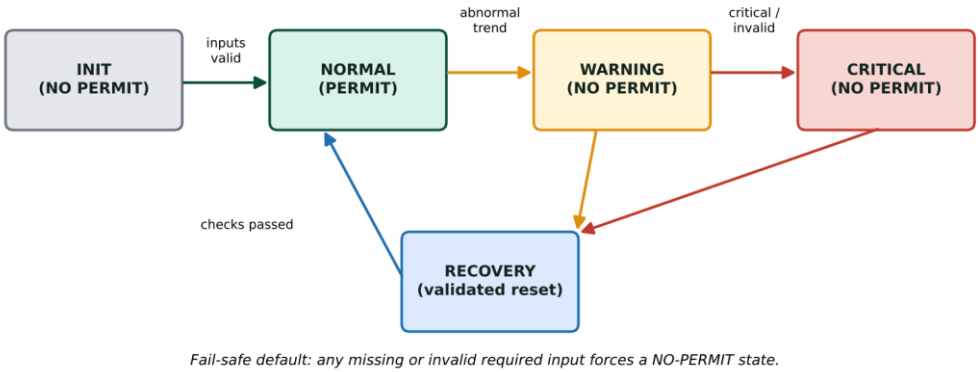


Figure 3.5: Safe-Drive Interlock finite-state machine

The permit is granted only in the NORMAL state. From WARNING or CRITICAL the system enters a RECOVERY state, and the permit is restored only after the faults have cleared and the inputs have been revalidated through a defined reset sequence.

3.13 CLOSED-LOOP PROTECTION AND SAFETY BLACK BOX

Depending on the severity and the diagnosis, the controller issues one or more protective actions: a buzzer / LED warning, a cooling request that switches the fan on, a controlled isolation through the protection relay, and the interlock block. Every event is written to the microSD card as a Safety Black Box record so that the sensor sequence, diagnosis, score, action and interlock state can be reviewed after the event. The recorded fields are given in Table 3.5.

Table 3.5: Safety Black Box record fields

Field	Description
Timestamp	Time of the sample or event
Sensor values	Cell voltages, pack current and four temperatures
Features	ΔV, ΔT, temperature rise rate, current trend
Safety Score	Score value and the top reasons for reduction
Diagnosis	Detected condition, suspected location and confidence

Field	Description
Action	Warning, cooling, isolation and interlock state
State	NORMAL / MONITOR / WARNING / CRITICAL and permit status

3.14 SOFTWARE-IN-THE-LOOP SIMULATOR

To validate the diagnostic workflow before and alongside hardware construction, a browser-based software-in-the-loop (SIL) simulator was developed. It reproduces the user interface and the decision flow using virtual sensor data, and no physical battery is connected. The simulator contains the following panels:

- **3D virtual battery and sensor network:** four cells (C1–C4), temperature sensors, the edge diagnostic board and an animated cooling fan.
- **Fault injection controls:** Normal, C3 Thermal, C2 Undervoltage, Overcurrent, Sensor Failure, Cooling Failure, Multiple Fault and Recover.
- **Real-time virtual telemetry:** pack voltage, pack current, maximum temperature and state of charge, with a voltage / temperature trend chart.
- **Explainable Safety Score:** a 0–100 gauge with the state (SAFE, WARNING or CRITICAL) and the reason.
- **Cell-level diagnostics and diagnostic explanation:** per-cell temperature and voltage, detected condition, suspected location, evidence and recommended action.
- **Closed-loop protection:** the drive permit (allowed / blocked), and the status of sensors, diagnostics, cooling and the interlock.
- **Safety Black Box:** a time-stamped event history of each injection, diagnosis and interlock decision.

In the simulator, each injected fault ramps the relevant virtual signals over five steps, reduces the Safety Score along a scripted profile, and then applies the final diagnosis, protective action and interlock block. The scripted score profile is a demonstration of the intended behaviour; the formula-based score of Section 3.10 is to be implemented and tuned in the ESP32 firmware.

3.15 DEVELOPMENT PLAN

Table 3.6: Development plan

Stage	Target work
Week 1	Finalize architecture, enclosure, sensor interfaces and safety boundaries
Week 2	ESP32 acquisition and local / dashboard visualization
Week 3	Diagnostic rules, Safety Score and fault localization
Week 4	Protection outputs, Safety Black Box and Safe-Drive Interlock
Week 5 onward	Edge anomaly model, repeatable validation scenarios, UI refinement and demonstration preparation

CHAPTER 4
RESULT

4.1 TEST METHOD AND SCOPE

The results in this chapter were obtained from the software-in-the-loop simulator described in Section 3.14, using controlled, simulated faults on virtual sensor data. The nominal condition of the virtual 4S pack is 16.08 V, 8.2 A, a maximum temperature of 32 °C, and a Safety Score of 94. Each fault is injected from the dashboard and the simulator is observed for five steps, after which the diagnosis, protective action and interlock state are recorded. The hardware prototype is under development; therefore, no result in this chapter should be read as a measurement on a physical battery.

4.2 SIMULATOR OUTPUT FOR REPRESENTATIVE SCENARIOS

Figure 4.1 shows the dashboard in the healthy condition. All inputs are valid, the Safety Score is 94, the state is SAFE and the drive permit is allowed.



Figure 4.1: Simulator dashboard in the normal (healthy) condition

Figure 4.2 shows the result of the C3 thermal fault. The temperature of cell C3 rises to 48 °C, the score falls to 50, the suspected location is identified as Cell 3 and the recommended action is “cooling on + drive interlock”. The drive permit is blocked, the interlock shows BLOCKED and the Black Box records the event.



Figure 4.2: Simulator dashboard after the C3 thermal fault injection

Figure 4.3 shows the multiple-fault scenario, in which temperature, current and voltage become abnormal together. The state becomes CRITICAL, the score falls to 30 and the maximum protection state is applied.



Figure 4.3: Simulator dashboard after the multiple-fault injection

4.3 SCENARIO-WISE RESULTS

Table 4.1 summarizes the final values of all scenarios. In every injected scenario the Safety Score decreased, the affected location was identified, the drive permit was blocked and the event was logged.

Table 4.1: Results of the simulated fault scenarios (value after step 5)

Scenario	Max temp (°C)	Current (A)	Pack voltage (V)	Score	State	Suspected location	Permit
Normal	32	8.2	16.08	94	SAFE	No fault	Allowed
C3 thermal	48	8.2	16.08	50	WARNING	Cell 3	Blocked
C2 undervoltage	32	8.2	14.68	56	WARNING	Cell 2	Blocked
Overcurrent	32	20.7	16.08	48	WARNING	Pack	Blocked
Sensor failure	32	8.2	16.08	30	CRITICAL	Sensor network	Blocked
Cooling failure	45.5	8.2	16.08	48	WARNING	Cooling subsystem	Blocked
Multiple fault	51	22.2	15.08	30	CRITICAL	C2 / C3 / pack	Blocked

In the sensor-failure scenario, the telemetry values are unchanged because the fault is an invalid or stale sensor signal; the freshness / plausibility check fails and the fail-safe rule blocks the permit. In the cooling-failure scenario, the temperature rises while the cooling status changes to FAIL, so no cooling action is available and the interlock blocks the permit.

4.4 SAFETY SCORE AND SIGNAL TRENDS

Figure 4.4 shows the fall of the Safety Score with each simulation step for the injected faults, together with the green / amber / red bands. The multiple-fault and sensor-failure profile falls fastest and enters the critical band, while single faults settle in the warning band. Figure 4.5 shows the corresponding ramps of temperature, current and voltage.

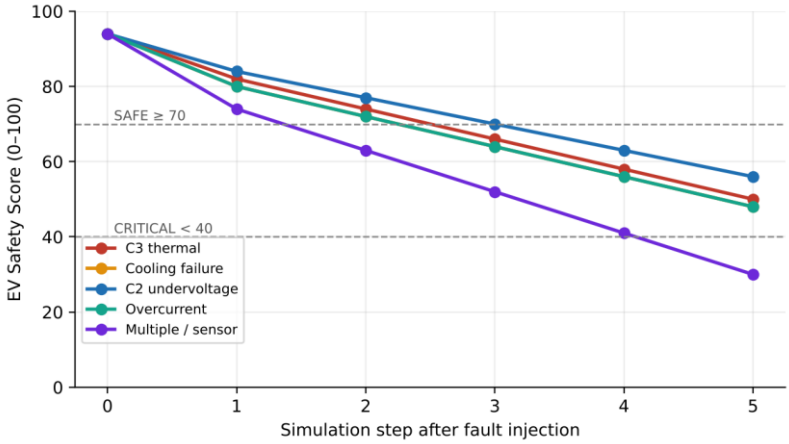


Figure 4.4: EV Safety Score versus simulation step for the injected faults

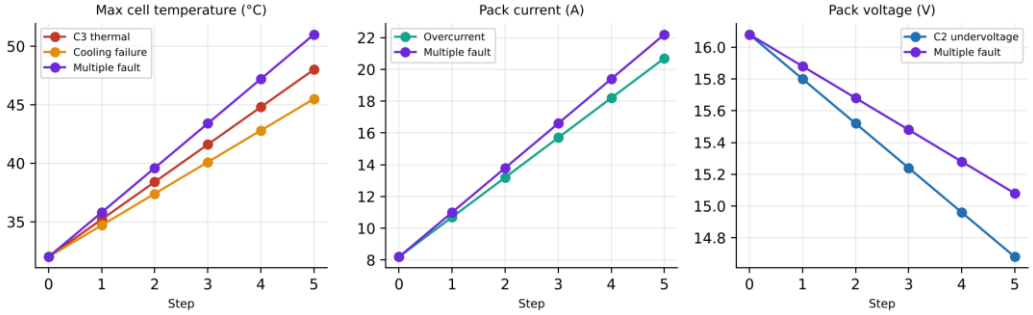


Figure 4.5: Virtual temperature, current and voltage trends during fault injection

4.5 INTERLOCK BEHAVIOUR AND SAFETY BLACK BOX

For every scenario the interlock changed from PERMIT to BLOCKED when the fault was applied. After the **Recover** command the simulator returned to the healthy state only after the inputs were revalidated and the permit was restored. The Safety Black Box recorded the sequence of events. A typical record sequence for the thermal fault is shown below.

```
SYSTEM INITIALIZED -> ALL INPUTS VALID -> DRIVE PERMIT
FAULT INJECTION STARTED: THERMAL
FAULT THERMAL -> DIAGNOSED -> PROTECTION -> INTERLOCK BLOCKED
FAULT CLEARED -> INPUTS REVALIDATED -> DRIVE PERMIT RESTORED
```

4.6 VALIDATION MATRIX FOR THE HARDWARE PROTOTYPE

The simulator demonstrates the intended behaviour of the logic. The same scenarios are to be repeated on the low-voltage hardware using safe test signals. Table 4.2 gives the validation matrix and its current status.

Table 4.2: Validation matrix and status

Test	Input	Expected state	Pass condition	Status
Normal operation	Stable voltage, current, temperature	Normal	Score high; monitoring continues	Simulated
Cell imbalance	One cell-voltage input shifted by test signal	Monitor / Warning	Imbalance flagged with suspected cell	Simulated (C2 undervoltage)
Temperature rise	Controlled increasing temperature trend	Warning	Risk increases; cooling recommended	Simulated
Critical fault	Combination of abnormal test signals	Critical	Alarm, protection output and log	Simulated
Unsafe drive request	Request with an invalid safety condition	Interlock	Activation blocked, warning, log	Simulated
Recovery	Signals returned to a valid state	Normal / Monitor	Cleared only after recovery checks	Simulated
All of the above on hardware	Safe test signals on the demonstrator	—	As above	Pending hardware build

4.7 DISCUSSION AND LIMITATIONS

- The simulated results show that combining several signals gives a more informative output than a single alarm: each scenario produced a condition, a location, evidence and an action.
- The Safety Score profile in the simulator is scripted; the weights of the formula in Section 3.10 must be tuned and validated against real measurements.
- Fault localization depends on sensor placement, measurement quality and the abnormal conditions represented during validation.
- A low-voltage demonstrator does not prove production EV performance or regulatory compliance, and the machine-learning layer must never be the sole safety mechanism.

CHAPTER 5
CONCLUSION AND FUTURE TRENDS

5.1 CONCLUSION

This project presented EV-GUARDIAN 2.0, a low-cost embedded diagnostic layer for an EV battery demonstrator that follows the sequence sense, diagnose, localize, predict, protect, interlock and record. The system integrates multi-parameter edge sensing, signal validation, fault localization, explainable diagnosis, a dynamic 0–100 Safety Score, automated protective response, a fail-safe Safe-Drive Interlock and a Safety Black Box in a single design.

The software-in-the-loop simulation of six controlled fault scenarios showed that the intended behaviour is achieved: the Safety Score fell, the affected cell or condition was identified, the drive permit was blocked and the event was recorded, and the permit returned only after the fault cleared and the inputs were revalidated. The demonstrator budget is estimated at about ₹4,400, which makes the concept suitable as an educational and demonstration platform. The hardware prototype is under development, and all further testing will use controlled simulated faults on low-voltage, current-limited hardware.

5.2 LIMITATIONS

- The Safety Score is an engineering indicator and not a certified safety metric.
- Fault localization depends on sensor placement and the conditions covered during validation.
- A low-voltage demonstrator does not prove production EV performance or compliance.
- Production use would require automotive functional-safety, EMC, battery, cybersecurity and regulatory validation.

5.3 FUTURE TRENDS

- Cell-level state-of-charge (SoC), state-of-health (SoH) and remaining-useful-life (RUL) estimation.
- Time-series anomaly detection trained on real or validated datasets.
- CAN integration with a real BMS / vehicle controller and authenticated diagnostics.
- Cloud-based fleet diagnostics and predictive-maintenance analytics.

- Digital-twin or physics-informed diagnostic models.
- Automotive-grade isolation, contactors and certified protection hardware.
- Cybersecurity monitoring and secure event logs.
- Extension to e-scooters, electric cars, buses, fleet monitoring and charging infrastructure.

5.4 ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS

By making battery risk understandable and supporting safer EV operation and maintenance, the project contributes to SDG 7 (affordable and clean energy), SDG 9 (industry, innovation and infrastructure), SDG 11 (sustainable cities and communities) and SDG 13 (climate action).

APPENDIX I
CORE LOGIC PSEUDOCODE

The following pseudocode summarizes the main control cycle and the Safe-Drive Interlock. It is a design outline and not the final firmware.

A. Main control cycle

```
loop every sample period:
    raw   = readSensors()            // V1-V4, I, T1-T4, fan, estop
    ok    = validate(raw)            // range, freshness, plausibility
    x     = filter(raw)              // moving average / smoothing
    f     = extractFeatures(x)       // dVcell, dT, dTdt, current
    rule  = evaluateRules(f)         // deterministic, with hysteresis
    anom  = anomalyScore(f)          // optional edge model
    loc   = localize(f, ok)          // suspected cell / condition
    risk  = sum(w[i] * R[i](f))      // normalized risk terms
    score = clamp(100 - risk, 0, 100)
    state = max(rule.severity, band(score))
    permit = interlock(state, ok, estop)   // see below
    act(state, loc, permit)          // buzzer, fan, relay, drive
    show(score, loc, reason); logBlackBox(...)
```

B. Safe-Drive Interlock

```
function interlock(state, ok, estop):
    if not allRequiredValid(ok): return NO_PERMIT  // fail-safe
    if estop == PRESSED:         return NO_PERMIT
    if state in (WARNING, CRITICAL): return NO_PERMIT
    if voltageBad or currentBad or tempTrendBad:
                                 return NO_PERMIT
    if inRecovery and not recoveryOK: return NO_PERMIT
    return PERMIT              // only when all checks pass
```

C. Simulator controls

The SIL simulator provides eight controls: Normal, C3 Thermal, C2 Undervoltage, Overcurrent, Sensor Failure, Cooling Failure, Multiple Fault and Recover. Each fault control starts the injection, updates the virtual telemetry for five steps, and applies the final diagnosis, protective action and interlock state.

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